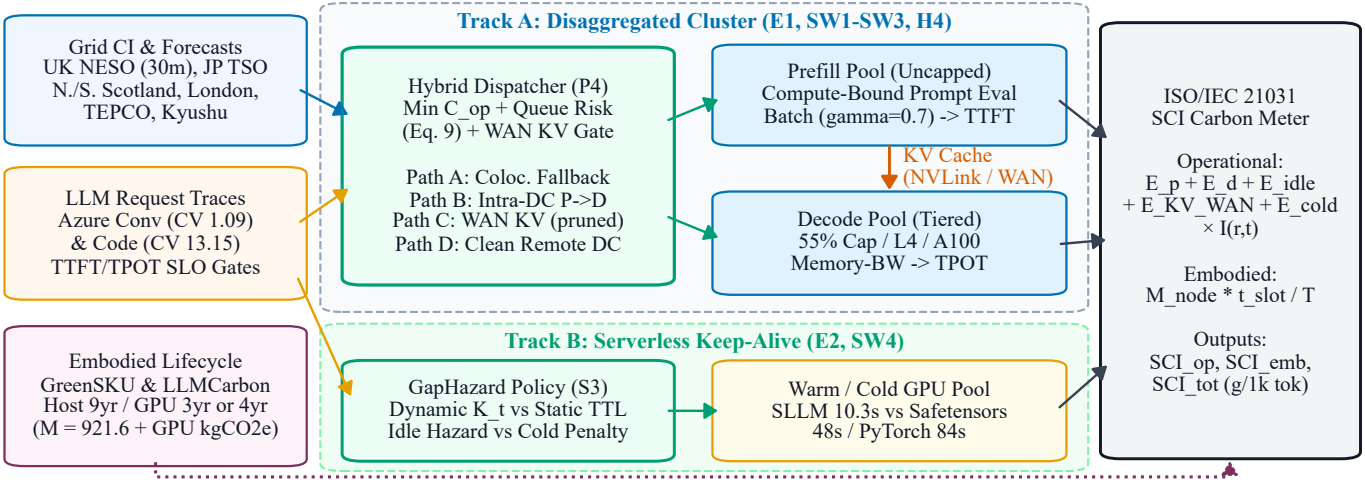

Grid CI & Forecasts
UK NESO (30m), JP TSO
N./S. Scotland, London,
TEPCO, Kyushu
LLM Request Traces
Azure Conv (CV 1.09)
& Code (CV 13.15)
TTFT/TPOT SLO Gates
Embodied Lifecycle
GreenSKU & LLMCarbon
Host 9yr / GPU 3yr or 4yr
(M = 921.6 + GPU kgCO2e)
Track A: Disaggregated Cluster (E1, SW1-SW3, H4)
Hybrid Dispatcher (P4)
Min C_op + Queue Risk
(Eq. 9) + WAN KV Gate
Path A: Coloc. Fallback
Path B: Intra-DC P->D
Path C: WAN KV (pruned)
Path D: Clean Remote DC
Prefill Pool (Uncapped)
Compute-Bound Prompt Eval
Batch (gamma=0.7) -> TTFT
KV Cache
(NVLink / WAN)
Decode Pool (Tiered)
55% Cap / L4 / A100
Memory-BW -> TPOT
Track B: Serverless Keep-Alive (E2, SW4)
GapHazard Policy (S3)
Dynamic K_t vs Static TTL
Idle Hazard vs Cold Penalty
Warm / Cold GPU Pool
SLLM 10.3s vs Safetensors
48s / PyTorch 84s
ISO/IEC 21031
SCI Carbon Meter
Operational:
E_p + E_d + E_idle
+ E_KV_WAN + E_cold
× I(r,t)
Embodied:
M_node * t_slot / T
Outputs:
SCI_op, SCI_emb,
SCI_tot (g/1k tok)